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Inventor(s)	Yang; Chunhui

TIRE MOUNTING COMPONENT AND DEVICE, AND TIRE MOUNTING/DISMOUNTING DEVICE

Abstract

Embodiments of the present disclosure provide a tire mounting component and device and a tire mounting/dismounting device. The tire mounting component comprises a main body and a pressing member capable of pressing down the tire, and the pressing member is arranged on the main body. A protective member is arranged on the main body, the protective member is at least partially located above the pressing member, and a portion of the protective member that is located above the pressing member can block at least a portion of a top surface of the pressing member. During mounting of the tire, the protective member can prevent an inner edge of the tire from coming into contact with the at least a portion of the top surface of the pressing member to prevent the inner edge of the tire from being stuck on the pressing member.

Inventors: **Yang; Chunhui (Mckinney, TX)**

Applicant: **Yang; Chunhui** (Mckinney, TX)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the field of tire mounting/dismounting devices, and in particular, to a component and tire mounting device, and a tire mounting/dismounting device.

BACKGROUND ART

[0002] When it is necessary to replace or repair tired, the tire needs to be mounted/dismounted. If the tire is mounted/dismounted manually, manual operations take a lot of time and require high labor intensity; and it is also difficult to control a manual force during mounting/dismounting, and the tire or a wheel disc is very likely to be damaged due to misoperation, thereby bringing about many disadvantages to people's safety when driving a vehicle. Therefore, an electric tire mounting/dismounting machine has appeared on the market, and electric mounting/dismounting of the tire is used instead of manual mounting/dismounting of the tire, which can improve the tire mounting/dismounting efficiency.

SUMMARY OF THE INVENTION

[0003] In order to overcome the shortcomings of the prior art, the present disclosure provides a component and tire mounting device, and a tire mounting/dismounting device.

[0004] The present disclosure is implemented by means of the following technical solutions.

[0005] In a first aspect, the present disclosure provides a tire mounting component, comprising a main body and a pressing member capable of pressing down the tire, wherein the pressing member is arranged on the main body; a protective member is arranged on the main body, the protective member is at least partially located above the pressing member, and a portion of the protective member that is located above the pressing member is capable of blocking at least a portion of a top surface of the pressing member; and during mounting of the tire, the protective member is capable of preventing an inner edge of the tire from coming into contact with the at least a portion of the top surface of the pressing member to prevent the inner edge of the tire from being stuck on the pressing member.

[0006] In a second aspect, the present disclosure provides a tire mounting device, comprising a driving device, a rotating structure, and a tire mounting structure, wherein the tire mounting structure comprises a tire mounting component as described above, the driving device is capable of driving the rotating structure to rotate, the tire mounting structure is arranged on the rotating structure, and the rotating structure is capable of driving the tire mounting structure to rotate.

[0007] In a third aspect, the present disclosure provides a tire mounting/dismounting device, comprising a tire dismounting structure and a tire mounting device as described above, wherein the tire dismounting structure comprises a supporting seat and a prying member detachably arranged on the supporting seat; the supporting seat comprises a second mounting portion, a second connecting portion, and a supporting arm, wherein the second mounting portion is arranged on the rotating structure, the second connecting portion is arranged on the second mounting portion, and the supporting arm is arranged on the second connecting portion; and the second connecting portion is provided with a retaining groove capable of transversely retaining the prying member, and baffles each having a length less than that of the supporting arm are arranged on two sides of a top of the supporting arm, and when the prying member is placed at a middle position between the baffles on the two side of the top of the supporting arm, the baffles on the two side of the top of the supporting arm are capable of transversely retaining the prying member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a schematic structural diagram of a tire mounting component according to an embodiment of the present disclosure;

[0009] FIG. **2** is a schematic structural diagram of a tire mounting structure according to an

embodiment of the present disclosure;
[0010] FIG. **3** is a schematic structural diagram of a tire mounting/dismounting device according to an embodiment of the present disclosure;
[0011] FIG. **4** is a schematic perspective view of a tire mounting/dismounting device according to an embodiment of the present disclosure;
[0012] FIG. **5** is a schematic structural diagram of a supporting seat according to an embodiment of the present disclosure;
[0013] FIG. **6** is a schematic structural diagram of a prying member according to an embodiment of the present disclosure; and
[0014] FIG. **7** is a schematic cross-sectional view of a head according to an embodiment of the present disclosure.
LIST OF REFERENCE SIGNS
[0015] **1**. Main body;
[0016] **2**. Pressing member;
[0017] **3**. Protective member;
[0018] **4**. Guide member;
[0019] **5**. Reinforcing rib;
[0020] **6**. Driving device;
[0021] **7**. Rotating structure;
[0022] **8**. Tire mounting structure;
[0023] **9**. First mounting portion;
[0024] **10**. First connecting portion;
[0025] **11**. Adjusting structure;
[0026] **12**. Supporting portion;
[0027] **13**. Adjusting member;
[0028] **14**. Housing;
[0029] **15**. Push rod;
[0030] **16**. Roller;
[0031] **17**. Handle;
[0032] **18**. Retaining member;
[0033] **19**. Mounting hole;
[0034] **20**. Electronic operation panel;
[0035] **21**. Tire dismounting structure;
[0036] **22**. Supporting seat;
[0037] **23**. Prying member;
[0038] **24**. Second mounting portion;
[0039] **25**. Second connecting portion;
[0040] **26**. Supporting arm;
[0041] **27**. Retaining groove;
[0042] **28**. Baffle;
[0043] **29**. Fixing structure;
[0044] **30**. Moving ring;
[0045] **31**. Fixing screw;
[0046] **32**. Head;
[0047] **33**. Elongated arm;
[0048] **34**. Handle.
DETAILED DESCRIPTION OF EMBODIMENTS
[0049] Specific embodiments of the present disclosure will be described in detail in this section. Preferred embodiments of the present disclosure are illustrated in the accompanying drawings. The accompanying drawings serve to supplement the text description of the specification with figures,

providing a visual understanding of each technical feature and the overall technical solution of the present disclosure, but cannot be construed as a limitation to the scope of protection of the present disclosure.
[0050] In the description of the present disclosure, it should be understood that orientation or position relationships indicated by terms “upper”, “lower”, “front”, “rear”, “left”, “right”, etc. are orientation or position relationships as shown in the accompanying drawings, and these terms are just used to facilitate description of the present disclosure and simplify the description, rather than indicating or implying that the mentioned device or element must have a specific orientation and must be constructed and operated in a specific orientation, and thus cannot be construed as a limitation to the present disclosure.
[0051] In the description of the present disclosure, “several” means one or more, “a plurality of” means two or more, “greater than”, “less than”, “over”, etc. are construed as excluding the number, and “above”, “below”, “within”, etc. are construed as including the number. The terms “first” and “second” in the description are merely intended to distinguish technical features, and cannot be construed as indicating or implying relative importance or implicitly indicating a number of the indicated technical features or implicitly indicating a sequence relationship of the indicated technical features.
[0052] In the description of the present disclosure, unless otherwise explicitly defined, the words such as “arrange”, “mount” and “connect” should be understood in a broad sense, and those skilled in the art can reasonably determine the specific meanings of the above words in the present disclosure with reference to the specific contents of the technical solutions.
[0053] Referring to FIG. **1**, a tire mounting component according to a preferred embodiment of the present disclosure comprises a main body **1** and a pressing member **2** capable of pressing down the tire. The pressing member **2** is arranged on the main body **1**. A protective member **3** is arranged on the main body **1**, the protective member **3** is entirely located above the pressing member **2**, and a portion of the protective member **3** that is located above the pressing member **2** can block an entire top surface of the pressing member **2**. During mounting of the tire, the protective member **3** can prevent an inner edge of the tire from coming into contact with the entire top surface of the pressing member **2** to prevent the inner edge of the tire from being stuck on the pressing member **2**.
[0054] In an embodiment, the protective member **3** is partially located above the main body **1**.
[0055] In an embodiment, the portion of the protective member **3** that is located above the main body **1** can block a portion of the top surface of the pressing member **2**, and during mounting of the tire, the protective member **3** can prevent the inner edge of the tire from coming into contact with the portion of the top surface of the pressing member **2** to prevent the inner edge of the tire from being stuck on the pressing member **2**. In the embodiment of the present disclosure, the shape of the main body **1** is not limited.
[0056] In the embodiment of the present disclosure, the portion of the protective member **3** that is located above the pressing member **2** can block the entire top surface or the portion of the top surface of the pressing member **2**, and during mounting of the tire, the protective member **3** can prevent the inner edge of the tire from coming into contact with the entire top surface or the portion of the top surface of the pressing member **2** to prevent the inner edge of the tire from being stuck on the pressing member **2**, thereby preventing the tire from being damaged due to the fact that the tire is stuck on the pressing member **2**, and reducing a degree of damage to the tire during mounting/dismounting of the tire.
[0057] In an optional embodiment, the pressing member **2** is a pressing wheel. The pressing wheel can roll during mounting of the tire, which reduces friction and makes the tire mounting process smoother. Moreover, the degree of damage to the tire can be reduced. In an embodiment, the pressing member **2** may be of other suitable structures, and is not limited thereto.
[0058] In the embodiment of the present disclosure, the specific structure of the protective member **3** is not limited, and may be configured according to actual requirements. In an optional

embodiment, one end of the protective member **3** close to the main body **1** is higher than the other end of the protective member **3** away from the main body **1**. An upper surface of the protective member **3** is configured as an outwardly protruding arc-shaped surface. In this way, when the inner edge of the tire comes into contact with the protective member **3**, the arc-shaped surface of the protective member **3** provides a guiding effect, so that the inner edge of the tire can slide along the arc-shaped surface of the protective member **3**, thereby preventing the inner edge of the tire from being stuck on the pressing member **2**, and reducing the degree of damage to the tire during mounting/dismounting of the tire. In an embodiment, the upper surface of the protective member **3** may be configured as an inclined surface.

[0059] In an optional embodiment, a guide member **4** capable of guiding the tire is arranged on the main body **1**, and a surface of the guide member **4** in contact with the inner edge of the tire is configured as an outwardly protruding arc-shaped surface. In this way, during mounting of the tire, the guide member **4** can fit with the inner edge of the tire well, thus greatly reducing wear of the inner edge of the tire by the guide member **4** and further reducing the degree of damage to the tire during mounting/dismounting of the tire. In an embodiment, the surface of the guide member **4** in contact with the inner edge of the tire may be a flat surface, or the surface of the guide member **4** in contact with the inner edge of the tire may be in other suitable shapes, and is not limited thereto.

[0060] In an optional embodiment, a reinforcing rib **5** is arranged on the main body **1**, and an end of the reinforcing rib **5** close to the pressing member **2** is bent and extends to the main body **1** close to the protective member **3**. In this way, the overall structural strength of the main body **1** is greatly improved, so that the tire mounting component according to the present disclosure is more durable and more reliable. In an embodiment, the reinforcing rib **5** may be configured in other suitable shapes, and is not limited thereto. In an embodiment, it is possible to provide no reinforcing rib **5** on the main body **1**.

[0061] Referring to FIGS. **2** to **4**, an embodiment of the present disclosure further provides a tire mounting device, comprising a driving device **6**, a rotating structure **7**, and a tire mounting structure **8**. The tire mounting structure **8** comprises a tire mounting component as described above, the driving device **6** can drive the rotating structure **7** to rotate, the tire mounting structure **8** is arranged on the rotating structure **7**, and the rotating structure **7** can drive the tire mounting structure **8** to rotate. With the above structure, during mounting of the tire, the driving device **6** can drive the rotating structure **7** to rotate, and the rotating structure **7** drives the tire mounting component to rotate, thus achieving a function of electrically mounting the tire. The above structure is simple, easy to mount and more convenient to use.

[0062] In the embodiment of the present disclosure, the tire mounting structure **8** is not limited, and may be configured according to actual requirements. Referring to FIG. **2**, in an optional embodiment, the tire mounting structure **8** comprises a first mounting portion **9**, a first connecting portion **10**, and an adjusting structure **11**. The first mounting portion **9** is arranged on the rotating structure **7**, the first connecting portion **10** is longitudinally movably arranged on the first mounting portion **9**, the tire mounting component is arranged on the first connecting portion **10**, and the adjusting structure **11** can adjust a positional height of the first connecting portion **10**. A supporting portion **12** is arranged on the first mounting portion **9**. The adjusting structure **11** comprises an adjusting member **13**. The adjusting member **13** is arranged on the first connecting portion **10**, and a bottom of the adjusting member **13** abuts against the supporting portion **12**. The first connecting portion **10** can move in a longitudinal direction of the adjusting member **13** and can be fixed at a current positional height after being adjusted. With the above structure, the positional height of the first connecting portion **10** can be adjusted according to actual requirements, and the positional height of the tire mounting component changes synchronously with the positional height of the first connecting portion **10**. When the tire mounting component is adjusted to an appropriate positional height, the process of mounting the tire is easier and more efficient. In addition, with the arrangement of the supporting portion **12**, the effect of supporting the adjusting member **13** is achieved, so that the first connecting portion **10** and the tire mounting component each have a more balanced and stable position, thereby making the mounting of the tire easier, more reliable, and more efficient. In an embodiment, it is possible to provide no supporting portion **12** on the first mounting portion **9**.

[0063] The specific structure of the adjusting structure **11** is not limited, and may be configured according to actual requirements. In an optional embodiment, the adjusting member **13** is provided with an external thread, the first connecting portion **10** is provided with a through hole having an internal thread, and the adjusting member **13** is arranged on the through hole through the cooperation of the external thread and the internal thread. With the above structure, since the first connecting portion **10** and the tire mounting component each have a certain weight, when the adjusting member **13** is rotated, the first connecting portion **10** and the tire mounting component do not move left and right. Under the cooperation of the external thread and the internal thread, the adjusting member **13** is rotated, so that the first connecting portion **10** and the tire mounting component move longitudinally along the adjusting member **13**. The above structure is simple to operate and has a high adjustment efficiency.

[0064] Referring to FIGS. **3** and **4**, in an optional embodiment, the tire mounting device further comprises a housing **14** and a push rod **15**. The push rod **15** is arranged on the housing **14**, rollers **16** are arranged at a bottom of the housing **14**, and the driving device **6**, the rotating structure **7** and the tire mounting structure **8** are all arranged on the housing **14**. An included angle between the push rod **15** and a horizontal plane is 30°-45°, preferably 35°, and the length of the push rod **15** is 50-90 cm, preferably 78 cm. A handle **17** perpendicular to the push rod **15** is arranged at an end of the push rod **15** away from the housing **14**, and the length of the handle **17** is 20-40 cm, preferably 25 cm. By using the above parameters, a worker can push the tire mounting device more easily and more efficiently while keeping his/her body upright. In an embodiment, the included angle between the push rod **15** and the horizontal plane may alternatively be a parameter other than 30°-45°, the length of the push rod **15** may be a parameter other than 50-90 cm, and the length of the handle **17** may be a parameter other than 20-40 cm.

[0065] In an optional embodiment, the housing **14** is provided with a retaining member **18** capable of fixing a hub and mounting holes **19** capable of mounting the retaining member **18**. One or more retaining members may be provided, and a plurality of mounting holes **19** may be provided. The retaining member **18** can be rotatably mounted on any mounting hole **19**. When it is necessary to fix the hub, the retaining member **18** is first mounted on a mounting hole **19** with a suitable size for the hub. After the hub is placed at a designated position, the retaining member **18** is rotated to retain the hub to prevent the hub from deviating, such that the tire can be mounted smoothly. With the above structure, the operation is easy, with a high efficiency and high stability. In an embodiment, other suitable retaining structures may be used instead of the above structure, and are not limited thereto.

[0066] In an optional embodiment, the housing **14** may be provided with a plurality of mounting positions, and the plurality of mounting positions are used to mount corresponding components. In this way, each component is stored more easily, and it is convenient to take and use the component. In an embodiment, the housing **14** may be provided with no mounting position described above.

[0067] In an optional embodiment, the housing **14** may be provided with an electronic operation panel **20**, the electronic operation panel **20** can control and adjust the driving device **6**, and the electronic operation panel **20** may also have other function buttons. In this way, the tire mounting device is easier to operate and more efficient. In an embodiment, it is possible to provide no electronic operation panel **20** on the housing **14**.

[0068] Referring to FIGS. **3** to **7**, an embodiment of the present disclosure further provides a tire mounting/dismounting device, comprising a tire dismounting structure **21** and a tire mounting device as described above. The tire dismounting structure **21** comprises a supporting seat **22** and a prying member **23**. The prying member **23** is detachably arranged on the supporting seat **22**. The supporting seat **22** comprises a second mounting portion **24**, a second connecting portion **25**, and a supporting arm **26**. The second mounting portion **24** is arranged on the rotating structure **7**, the second connecting portion **25** is arranged on the second mounting portion **24**, and the supporting arm **26** is arranged on the second connecting portion **25**. The second connecting portion **25** is provided with a retaining groove **27** capable of transversely retaining the prying member **23**, and baffles **28** each having a length less than that of the supporting arm **26** are arranged on two sides of a top of the supporting arm **26**. When the prying member **23** is placed at a middle position between the baffles **28** on the two side of the top of the supporting arm **26**, the baffles **28** on the two side of the top of the supporting arm **26** can transversely retain the prying member **23**. With the above structure, since the prying member **23** needs to be inserted between the inner edge of the tire and the hub during dismounting of the tire, and then rotates along the inner edge of the tire to separate the tire from the hub, the placement of the prying member **23** in the retaining groove **27** and at the middle position between the baffles **28** on the two sides of the top of the supporting arm **26** can well prevent the prying member **23** from loosening and swinging. This can protect the worker and prevent the worker from being injured due to the loosening and swinging of the prying member **23** during dismounting of the tire, and can also make the dismounting of the tire easier and more efficient.

[0069] In the embodiment of the present disclosure, the specific structure of the tire dismounting structure **21** is not limited, and may be configured according to actual requirements. Referring to FIG. **5**, in an optional embodiment, the tire dismounting structure **21** comprises a fixing structure **29**. When it is necessary to adjust a positional height of the supporting seat **22**, the fixing structure **29** is loosened, and the supporting seat **22** can move longitudinally. The fixing structure **29** can fix the supporting seat **22** after the positional height of the supporting seat **22** is adjusted. With the above structure, the positional height of the supporting seat **22** can be adjusted according to actual requirements, and thus the tire dismounting process is easier and more efficient. In an embodiment, the tire dismounting structure **21** may comprise no fixing structure **29**.

[0070] Further, in the embodiment of the present disclosure, the specific structure of the fixing structure **29** is not limited, and may be configured according to actual requirements. In an optional embodiment, the fixing structure **29** comprises a moving ring **30** and a fixing screw **31**. The moving ring **30** is sleeved on the second mounting portion **24**, and the moving ring **30** can move longitudinally on the second mounting portion **24**. The fixing screw **31** runs through a side wall of the moving ring **30**. When it is necessary to adjust the positional height of the supporting seat **22**, the fixing screw **31** is rotated, such that the fixing screw **31** is separated from the second mounting portion **24**, and thus the positional height of the supporting seat **22** can be adjusted. After the positional height of the supporting seat **22** is adjusted, the moving ring **30** abuts against the rotating structure **7**, and the fixing screw **31** is rotated in an opposite direction, such that the fixing screw **31** abuts against the second mounting portion **24**, and thus the supporting seat **22** can be fixed. In an embodiment, it is possible to dismount the supporting portion **12** of the first mounting portion **9**, mount the supporting portion **12** on the rotating structure **7**, and mount the second mounting portion **24** on the supporting portion **12**. After the positional height of the supporting seat **22** is adjusted, the moving ring **30** abuts against the supporting portion **12**, and the fixing screw **31** is rotated in the opposite direction, such that the fixing screw **31** abuts against the second mounting portion **24**, and thus the supporting seat **22** can be fixed. In an embodiment, the fixing structure **29** may alternatively be other suitable structures.

[0071] In the embodiment of the present disclosure, the specific structure of the prying member **23** is not limited, and may be configured according to actual requirements. Referring to FIGS. **6** and **7**, in an optional embodiment, the prying member **23** comprises a head **32** and an elongated arm **33**. The head **32** is connected to the elongated arm **33**. A surface of the head **32** that abuts against the tire when dismounting the tire is configured as an arc-shaped surface. Since the prying member **23** needs to be inserted between the inner edge of the tire and the hub during dismounting of the tire, and then rotates along the inner edge of the tire to separate the tire from the hub, the surface of the head **32** that abuts against the tire when dismounting the tire is configured as an arc-shaped surface. In this way, the damage to the tire caused by the prying member **23** during dismounting of the tire can be reduced, thereby achieving better safety. In an embodiment, the surface of the head **32** that abuts against the tire when dismounting the tire may be configured as a flat surface, or the surface of the head **32** that abuts against the tire when dismounting the tire may be configured in other shapes, and is not limited thereto.

[0072] Referring to FIG. **6**, in an optional embodiment, two sides of an end of the head **32** away from the elongated arm **33** are configured in the shape of an outwardly protruding arc. The prying member **23** needs to be inserted between the inner edge of the tire and the hub during dismounting of the tire, and then rotates along the inner edge of the tire to separate the tire from the hub. Therefore, configuring the two sides of the end of the head away from the elongated arm **33** in the shape of an outwardly protruding arc can reduce the damage to the inner edge of the tire and the hub during the insertion of the prying member **23** between the inner edge of the tire and the hub, thereby achieving better safety. In an embodiment, the two sides of the end of the head **32** away from the elongated arm **33** may be in the shape of a corner, or the two sides of the end of the head **32** away from the elongated arm **33** may be configured in other shapes, and is not limited thereto.

[0073] In an optional embodiment, a plurality of edges and corners are arranged at a joint between the head **32** and the elongated arm **33**, and the size of the joint between the head **32** and the elongated arm **33** is gradually increased from the elongated arm **33** to the head **32**. With the above structure, the prying member **23** has a higher overall strength, and is more reliable during dismounting of the tire, thereby achieving higher safety. In an embodiment, no edges and corners may be provided at the joint between the head **32** and the elongated arm **33**, or the size of the joint between the head **32** and the elongated arm **33** is constant or gradually reduced from the elongated arm **33** to the head **32**, or other suitable structures may be used at the joint between the head **32** and the elongated arm **33**, and are not limited thereto.

[0074] In an optional embodiment, the prying member **23** is of an integrally formed structure. In this way, the prying member **23** has a higher overall strength, and is more reliable during dismounting of the tire, thereby achieving higher safety. In an embodiment, the prying member **23** may be formed by welding two or more components, the prying member **23** may be formed by assembling two or more components, or the prying member **23** may be of other suitable structures, and is not limited thereto.

[0075] In an optional embodiment, a handle **34** is arranged at an end of the elongated arm **33** away from the head **32**, and the handle **34** is made of a rubber material. In this way, the worker can use the prying member **23** to dismount the tire more easily, and the occurrence of injuries to the worker can be reduced, thereby achieving higher safety. In an embodiment, it is possible to use the handle **34** made of other materials instead of the handle **34** made of the rubber material, or it is possible to provide no handle **34**.

[0076] A tire mounting workflow of the tire mounting/dismounting device according to an embodiment of the present disclosure may be as follows.

[0077] 1. The hub is first placed at a designated position above the housing **14**, to facilitate fixing of the hub by means of the retaining member **18**.

[0078] 2. The tire mounting structure **8** is mounted on the rotating structure **7**, and the positional height of the tire mounting component is adjusted by using the adjusting structure **11**, so that the positional height of the tire mounting component matches the size of the hub.

[0079] 3. Lubricating oil is applied to an inner edge of a tire to be mounted.

[0080] 4. The driving device **6** is started, and the rotating structure **7** drives the tire mounting component to rotate.

[0081] 5. The tire is placed obliquely, so that a lower inner edge portion of the tire is placed on the main body **1** and is in contact with the guide member **4**. Under the action of the rotation of the tire

mounting component, the guide member **4** moves along the lower inner edge of the tire, waiting for the mounting of the lower inner edge of the tire on the hub.
[0082] 6. A side of the tire is pressed downward, so that the positional height of the pressing member **2** exceeds the positional height of a side of an upper inner edge of the tire. In this process, due to the arrangement of the protective member **3**, the protective member **3** can preferentially be in contact with the upper inner edge of the tire to prevent the upper inner edge of the tire from being stuck on the pressing member **2**, thereby preventing the tire from being damaged due to the fact that the tire is stuck on the pressing member **2**.
[0083] 7. The pressing member **2** is pressed on the side of the upper inner edge of the tire, and the rotating structure **7** drives the tire mounting component to rotate. The guide member **4** moves along the upper inner edge of the tire, and the pressing member **2** can move along the upper inner edge to gradually press the entire upper inner edge of the tire downward, waiting for the mounting of the upper inner edge of the tire on the hub, thus completing the mounting of the tire.
[0084] A tire dismounting workflow of the tire mounting/dismounting device according to an embodiment of the present disclosure may be as follows.
[0085] 1. A tire to be dismounted and the hub are placed at a designated position above the housing **14**, to facilitate fixing of the hub by means of the retaining member **18**.
[0086] 2. The supporting portion **12** of the first mounting portion **9** is dismounted and mounted on the rotating structure **7**, the supporting seat **22** is mounted on the supporting portion **12**, and the positional height of the supporting seat **22** is adjusted by using the fixing structure **29**, so that the positional height of the supporting seat **22** matches the size of the tire and the size of the hub.
[0087] 3. Lubricating oil is applied on the arc-shaped surface of the prying member **23**.
[0088] 4. The head **32** of the prying member **23** is inserted between an upper inner edge of the tire and the hub, and then pries up, such that the elongated arm **33** of the prying member **23** is placed in the retaining groove **27** of the supporting seat **22** and between the baffles **28** on the two sides of the top of the supporting arm **26**. In this case, the arc-shaped surface of the head **32** of the prying member **23** abuts against an inner wall of the tire.
[0089] 5. The handle **34** is manually grasped to rotate the prying member **23**, such that the prying member **23** moves along the upper inner edge of the tire, and the prying member **23** gradually pries up the entire upper inner edge of the tire.
[0090] 6. The prying member **23** is taken out, and the head **32** of the prying member **23** is inserted between a lower inner edge of the tire and the hub, and then pries up, such that the elongated arm **33** of the prying member **23** is placed in the retaining groove **27** of the supporting seat **22** and between the baffles **28** on the two sides of the top of the supporting arm **26**. In this case, the arc-shaped surface of the head **32** of the prying member **23** abuts against an outer wall of the tire.
[0091] 7. The handle **34** is manually grasped to rotate the prying member **23**, such that the prying member **23** moves along the lower inner edge of the tire, and the prying member **23** gradually pries up the entire lower inner edge of the tire, thus completing the dismounting of the tire.
[0092] Those skilled in the art can freely combine and use the above additional technical features provided that no conflict occurs.
[0093] The foregoing implementations are only preferred implementations of the present disclosure and cannot be used to limit the scope of protection of the present disclosure. All non-essential modifications and substitutions made by those skilled in the art on the basis of the present disclosure shall fall within the scope of protection of the present disclosure.

Claims

1. A tire mounting component, comprising a main body with a flat top and rounded sides and a pressing wheel capable of pressing down the tire, wherein the pressing wheel is arranged on one side of the flat main body; a protective member is arranged on the one side of the main body, the protective member is at least partially located above the pressing wheel, wherein the top of the protective member has an arc-shaped surface and one end of the protective member that is closer to the main body has a greater dimension in a direction extending away from the pressing wheel than the other end of the protective member that is further away from the main body; the protective member prevents the inner edge of the tire from coming into contact with the at least a portion of the top surface of the pressing wheel based on the inner edge of the tire sliding along the arc-shaped top of the protective member;

2. (canceled)

3. The tire mounting component according to claim 1, wherein an arc-shaped guide member on the flat top of the main body is configured to contact with inner edge of the tire as the tire sits on the top of the flat main body.

4. The tire mounting component according to claim 3, wherein a reinforcing rib is arranged on the flat top of the main body, an end of the reinforcing rib close to the pressing wheel, is curved and extends from the protective member to the arc-shaped guide member

5. (canceled)

6. (canceled)

7. (canceled)

8. (canceled)

9. (canceled)

10. (canceled)
